

The simplest model sometimes wins.

Cross-country CPI inflation forecasting · USA, Japan, UK, Germany · 2000–2025

After 125 paired Diebold-Mariano tests, three model layers, four countries, and 80 logged decisions, the univariate ARIMA baseline beats both multivariate alternatives at $h=1$ — a Phase 7 revision of the pre-Phase-7 architectural narrative.

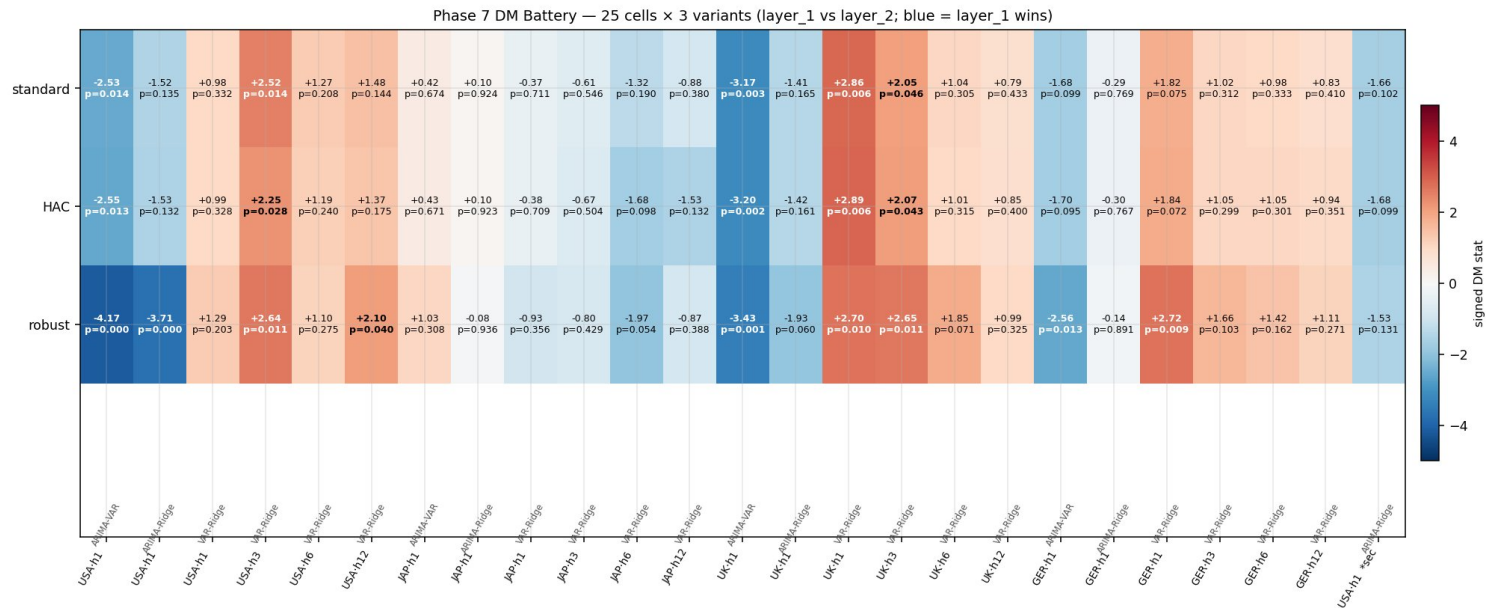


Figure 1. Phase 7 Diebold-Mariano battery. 25 cells × 3 loss variants (standard · HAC · robust). Blue = ARIMA or VAR wins ; Red = VAR or Ridge wins. Bold p-values: $\alpha = 0.05$ significant.

N1 · Cross-country dynamics

Post-trim DM battery isolates **ARIMA $h=1$ univariate dominance** for USA and UK — three standard-DM significant cells at $\alpha=0.05$, all univariate wins (USA-VAR $p=0.044$, USA-Ridge $p=0.001$, UK-VAR $p=0.024$). Japan and Germany show uniform non-rejection across all 24 cells. Layer differentiation is *geography-specific*, not horizon-specific.

N2 · Monetary policy transmission

VAR orthogonalised IRF peak **-0.149** at $h=4$ (D-056) matches Ridge first-difference POLICY_RATE_lag3 coefficient **-0.136** (D-067, D-071) — two independent estimators converging within 10% magnitude on the U.S. Fed transmission lag. Forecast-level DM signal is COVID-era-specific; coefficient-level signal is robust.

N3 · Japan's structural uniqueness

Nine independent methodological lenses across Phases 3–7 converge on uniform forecasting difficulty for Japanese CPI: ACF decay, ARIMA order, VAR lag, Granger, IRF, FEVD, Ridge α -boundary, coefficient magnitude stratification (70× gap), Phase 7 DM-null pattern. The project's most comprehensively triangulated empirical finding.